

Direction, Ballisticity, and the Zero-Speed Chamber for One-Dimensional IID Random Environment

HCS-C348 source-local reconstruction

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Abstract

We give a convention-complete reconstruction of the one-dimensional iid nearest-neighbour random walk in random environment. One theorem joins the quenched finite-interval scale function, the log-bias direction trichotomy, the annealed almost-sure velocity, and the full transient zero-speed region. The latter is proved analytically: stationary ergodic crossing times are truncated before Birkhoff's theorem, forcing infinite passage time per unit distance and hence zero speed. Beta and constant environments close all stated parameter faces. Exact finite ledgers are regression receipts only. No priority, arithmetic-target, Euler-factor, root-number, target-zero, or Hilbert–Pólya claim is made.

Revision certificate. Scale and direction closure.

1 Frozen model and the two laws

Let $(\Omega, \mathcal{F}, \mathbf{P})$ carry iid variables $\omega = (\omega_x)_{x \in \mathbb{Z}}$ such that

$$0 < \omega_x < 1, \quad \rho_x = \frac{1 - \omega_x}{\omega_x}, \quad \mathbf{E}|\log \rho_0| < \infty. \quad (1)$$

Given the complete frozen environment, the quenched law P_ω^z is the Markov law started at z with

$$P_\omega(X_{n+1} = x + 1 \mid X_n = x) = \omega_x, \quad P_\omega(X_{n+1} = x - 1 \mid X_n = x) = 1 - \omega_x. \quad (2)$$

The annealed law is $\mathbb{P}^z(d\omega, dX) = \mathbf{P}(d\omega)P_\omega^z(dX)$. Thus the medium is sampled once; annealing does not resample ω_x on a revisit. Put $T_y = \inf\{n \geq 0 : X_n = y\}$ and $m = \mathbf{E} \log \rho_0$.

Theorem 1 (Complete phase theorem). *Under (1) the following statements hold.*

(i) *For integers $a < x < b$, define $R_a = 1$ and $R_k = \prod_{j=a+1}^k \rho_j$. Then*

$$P_\omega^x(T_b < T_a) = \frac{\sum_{k=a}^{x-1} R_k}{\sum_{k=a}^{b-1} R_k}. \quad (3)$$

(ii) *For $\mathbf{P}$ -almost every ω , the following holds P_ω^0 -almost surely:*

$$\begin{array}{l|l} m < 0 & X_n \rightarrow +\infty, \\ m = 0 & \liminf X_n = -\infty \text{ and } \limsup X_n = +\infty, \\ m > 0 & X_n \rightarrow -\infty. \end{array} \quad (4)$$

(iii) *Under $\mathbb{P}^0$, and therefore under P_ω^0 for $\mathbf{P}$ -almost every environment,*

$$\frac{X_n}{n} \rightarrow v = \begin{cases} \frac{1 - \mathbf{E}\rho_0}{1 + \mathbf{E}\rho_0}, & \mathbf{E}\rho_0 < 1, \\ -\frac{1 - \mathbf{E}\rho_0^{-1}}{1 + \mathbf{E}\rho_0^{-1}}, & \mathbf{E}\rho_0^{-1} < 1, \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

The two nonzero cases are disjoint.

2 Scale and direction

Proof of Theorem 1(i). Let $h(i) = P_\omega^i(T_b < T_a)$. It is the unique solution of

$$h(i) = \omega_i h(i+1) + (1 - \omega_i) h(i-1), \quad h(a) = 0, h(b) = 1.$$

For $\Delta_i = h(i) - h(i-1)$, this becomes $\Delta_{i+1} = \rho_i \Delta_i$. Hence $\Delta_{k+1} = R_k \Delta_{a+1}$. Summing and imposing the two boundary values gives (3); positivity of the denominator gives uniqueness. $\square$

For the infinite line define the two-sided potential

$$V(0) = 0, \quad V(n) = \sum_{j=1}^n \log \rho_j \quad (n > 0), \quad V(n) = - \sum_{j=n+1}^0 \log \rho_j \quad (n < 0). \quad (6)$$

The strong law gives $V(n)/n \rightarrow m$ at both ends. If $m < 0$, scale weights decay to the right and grow to the left. Sending the barriers in (3) to infinity shows that every positive level is hit and every fixed level is visited only finitely often; thus $X_n \rightarrow +\infty$. Reflection gives the case $m > 0$. If $m = 0$, the iid sums in (6) oscillate at both ends unless they vanish identically; the latter is the simple symmetric environment. Both one-sided scale sums diverge in either case, and (3) shows that every integer is hit. This proves (4).

3 Source and ownership note

Solomon's primary paper [1] owns the classical iid direction and speed theorem. Zeitouni's review [2] supplies an authoritative later formulation. Our contribution is a source-local, auditable reconstruction joining the proof, exact regression receipt, boundary atlas, and Route-A evaluation. We make no priority claim.

References

- [1] F. Solomon, Random walks in a random environment, *Ann. Probab.* **3** (1975), 1–31. <https://doi.org/10.1214/AOP/1176996444>.
- [2] O. Zeitouni, Random walks in random environment, in *Lectures on Probability Theory and Statistics*, Lecture Notes in Mathematics 1837, Springer (2004), 189–312. https://doi.org/10.1007/978-3-540-39874-5_2.